

Signals and Systems

Prof. Xu (Judy) Zhu

Faculty of Informatics

Harbin Institute of Technology (Shenzhen)

This course is a professional foundation course that examines the basic concepts and basic analytical methods of signals and systems. Preliminary understanding of how to establish a **mathematical model** of signals and systems, and after mathematical analysis and solution, the results are physically explained and given **physical meaning**.

Textbook

Simon Haykin and Barry Van Veen, Signals and Systems (Second Edition), John Wiley & Sons, 2002, 9780471164746.

Reference book

Alan V. Oppenheim, Signals and Systems (Second Edition), Prentice-Hall, 1996.

Why Study Signals & Systems in the AI Era?

DeepSeek: Signals & Systems (S&S) is the Foundational Language Connecting Physics to AI.

1. AI's Data is Signals

- **Computer Vision:** Images = 2D Spatial Signals
- **Speech/Audio:** Sound = 1D Time-Based Signals
- **Sensor Data (IoT, AV):** LiDAR, Radar = Continuous Signal Streams
- **S&S Provides the Tools** to represent, analyze, and process this real-world data.

2. It's the Math Behind Core AI Architectures

- **CNNs (Convolutional Neural Networks):** Their core operation, **convolution**, is a fundamental S&S concept for filtering and feature extraction.
- **RNNs/LSTMs:** Process sequential data (time-series signals) using concepts of system memory and feedback.

3. The Critical Analog-to-Digital Bridge

- **Sampling Theorem (Nyquist):** Prevents **aliasing** (data distortion) when converting real-world analog data for AI models.
- **Digital Filter Design:** Essential for preprocessing data (noise removal, feature enhancement) before it reaches the model.

4. System-Level Thinking for Real-World AI

- **Beyond the Model:** AI is part of a larger **system** (e.g., a robot, a self-driving car).
- **Control Theory & Feedback:** S&S provides concepts for stability, response time, and latency crucial for real-time, reliable deployment.

Without S&S: You are an **API User**. You can apply tools but may struggle to debug, innovate, or handle complex, real-world data effectively.

With S&S: You become a **System Architect**. You understand *why* models work, can design better systems, and push the boundaries of what's possible.

It's not outdated—it's essential for building robust, efficient, and innovative AI.

Assessment items	Percentage	Description
Final Exam	80%	Scope of assessment: all contents. Closed-book exam.
Class Participation	20%	The score is calculated according to the percentage of the number of answers to the number of questions initiated by the teacher to participate in the <i>Rain Class</i> , regardless of whether it is right or wrong.

Contents of this lecture

1.1 Introduction

1.2 Signal classification and typical signals

1.3 Signal operation

1.4 Singular signals

Goals of this lecture

1. Understand how signals are classified
2. Understand the characteristics and physical significance of typical signals;
3. Proficient in signal calculation, be able to correctly draw the waveform after time shift, reverse fold and scale transformation step by step;
4. Familiarize yourself with the properties, effects, and interrelationships of exotic signals, especially step and impulse functions.

Message: A symbol to be transmitted in a way agreed in advance by both the sender and receiver, such as language, text, images, data, etc.

Information: Unknown content obtained from the received message.

Signal: A signal is a **variation** of a measurable quantity that carries **information** about the behaviour of a system. Examples: human voice, room temperature controlled by a thermostat system, position, speed, and acceleration of an aircraft, force measured with force sensors in robotic systems, digital photographs, music recording etc.

Electrical signal: The current or voltage that changes corresponding to the message (language, text, image, data), or the charge on the capacitor, the magnetic flux in the inductor, etc.

Signal Processing

- We interpret signals to make **decisions**.
- Signals allow us to see, hear, feel and act.
- Signal analysis allows us to **understand** or interpret an aspect of a signal which is *significant but not obvious on first inspection*.
- **Signal processing** is the science of analyzing, modifying, or synthesizing signals.

Please name at least 2 applications of signal processing.

1.1 Introduction

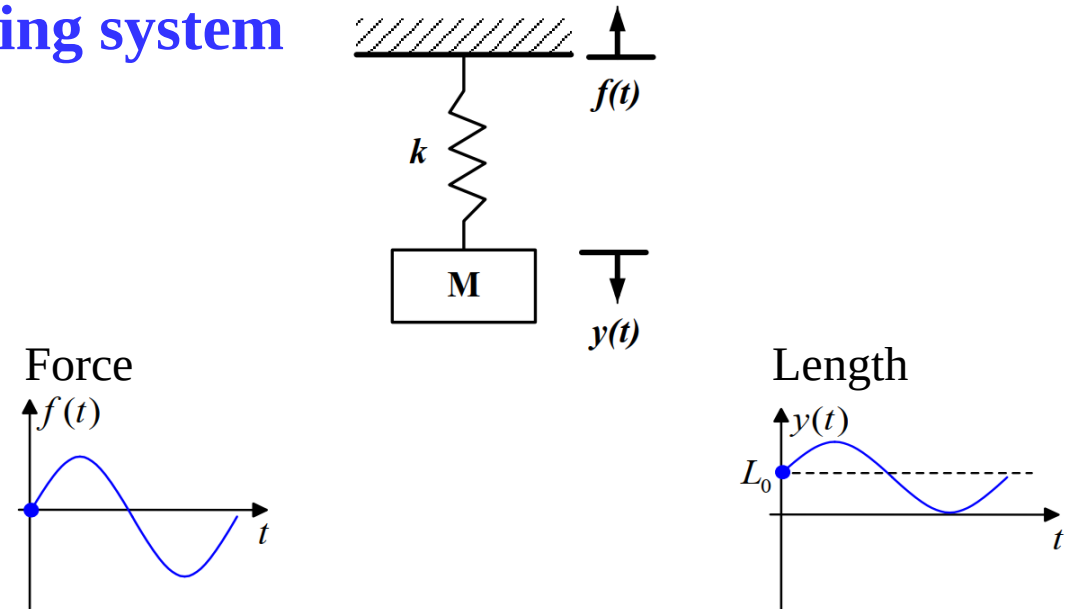


System: A set of things that are connected to each other in a specific function.

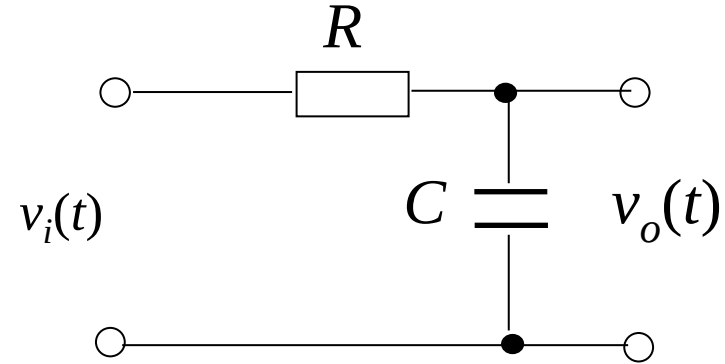


- **Physical systems:** Circuit systems, communication systems, automatic control systems, mechanical systems, optical systems, etc.;
- **Non-physical systems:** biological systems, political system systems, economic structure systems, transportation systems, meteorological systems, etc.

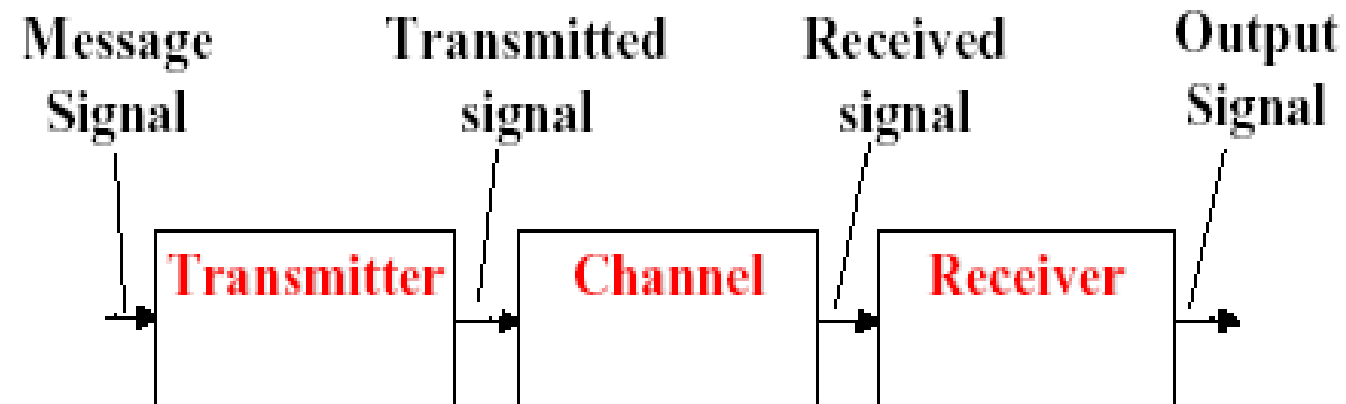
Spring system



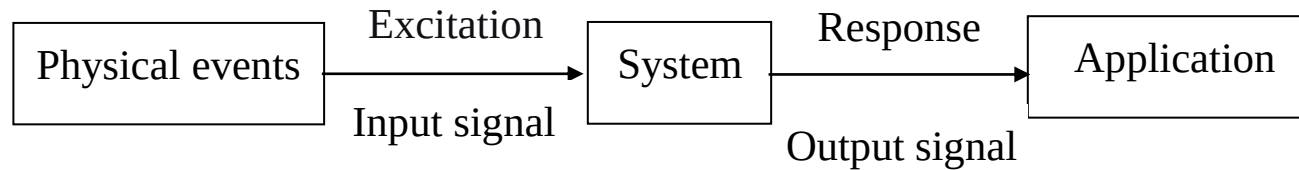
- **Electrical Circuit System**



- **Communication system**



1.1 Introduction



For a system, the input signal is often called **excitation**, and the output signal is often called the **response**.

Problems solved by this course in Signals and Systems:

- 1) What kind of response does the incentive system produce?
- 2) What kind of system is designed for a given excitation and required response?
- 3) What kind of incentives are needed to facilitate system analysis and the required response?

Excitation + system

→ response

Excitation → system

← response

Excitation ←

System + Response

Note: Signals and systems are closely related and complement each other. Without the signal, the system loses its meaning.

1.2.1 Classification of Signals

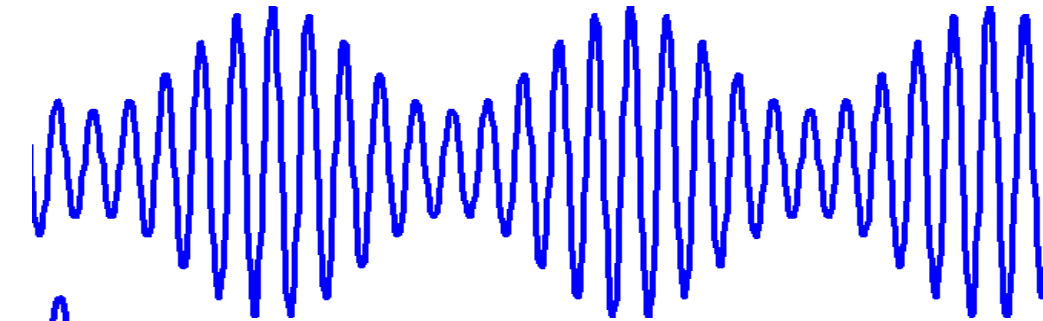
1. Deterministic Signals and Random Signals

Deterministic signal - for a certain moment, the signal has a definite value corresponding to it.

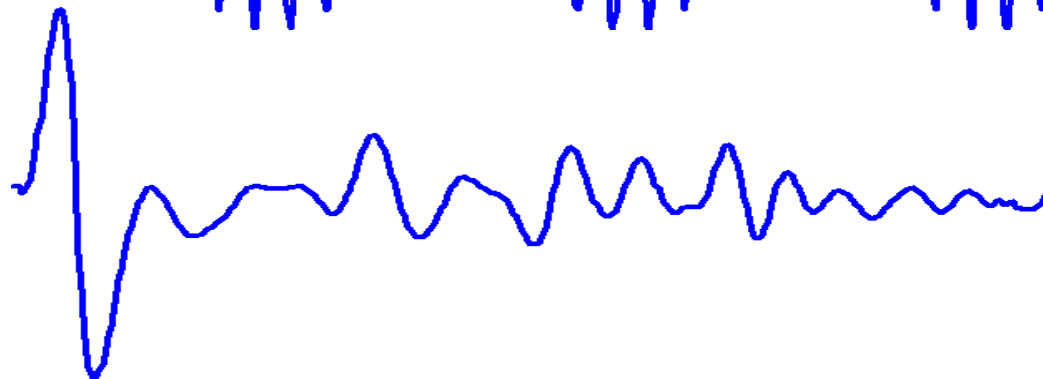
Random signals - unpredictable signals such as noise.

Deterministic signals

$$S(t) = (2 + \cos \omega t) \cos (10 \omega t)$$



Random signal



2. Periodic Signals and Non-Periodic Signals

Periodic signal: It starts and starts at a certain time interval, and it is a signal without beginning or end. A signal with a basic period of T - $f(t)=f(t+T)$ for all t .

Non-periodic signal: A signal that does not meet the characteristics of the cycle in time.

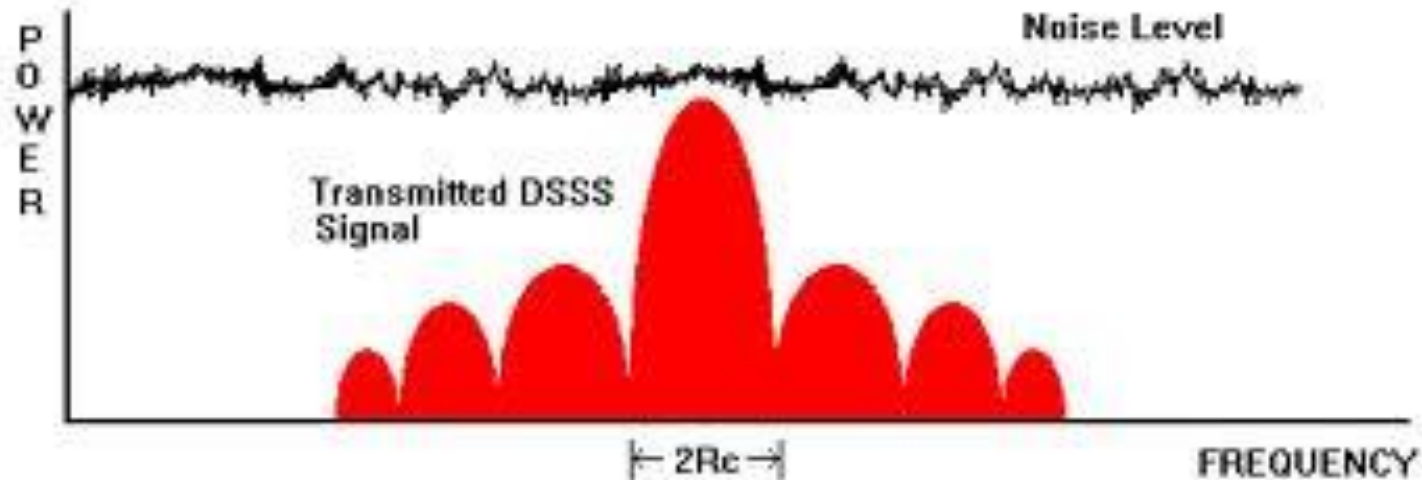
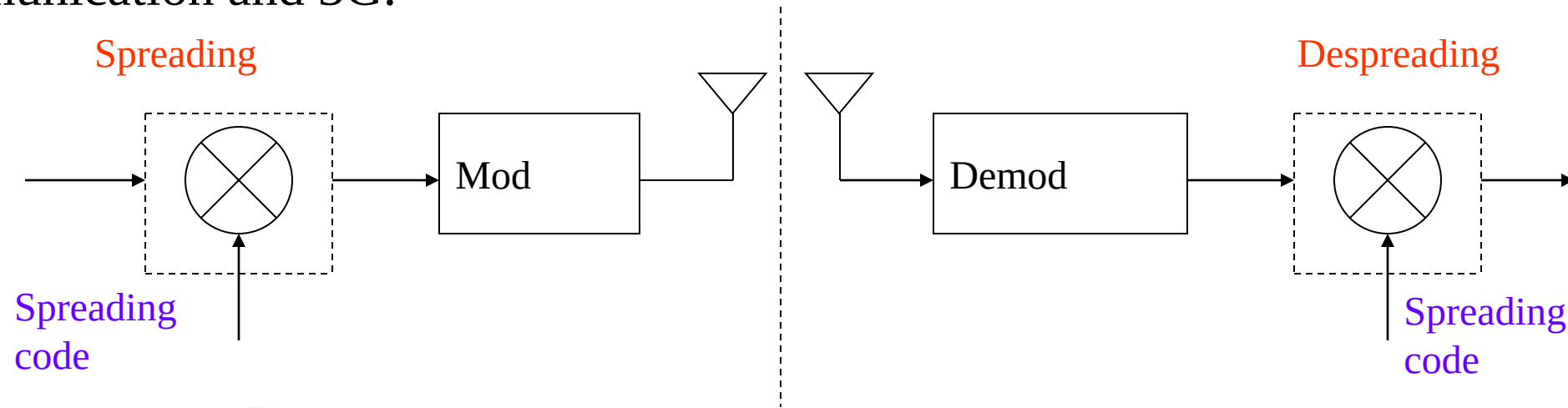
The spreading code of direct-sequence spread spectrum is

- ☒ A A periodic signal
- ☐ B A non-periodic signal

Submit

1.2 Signal Classification and Typical Signals

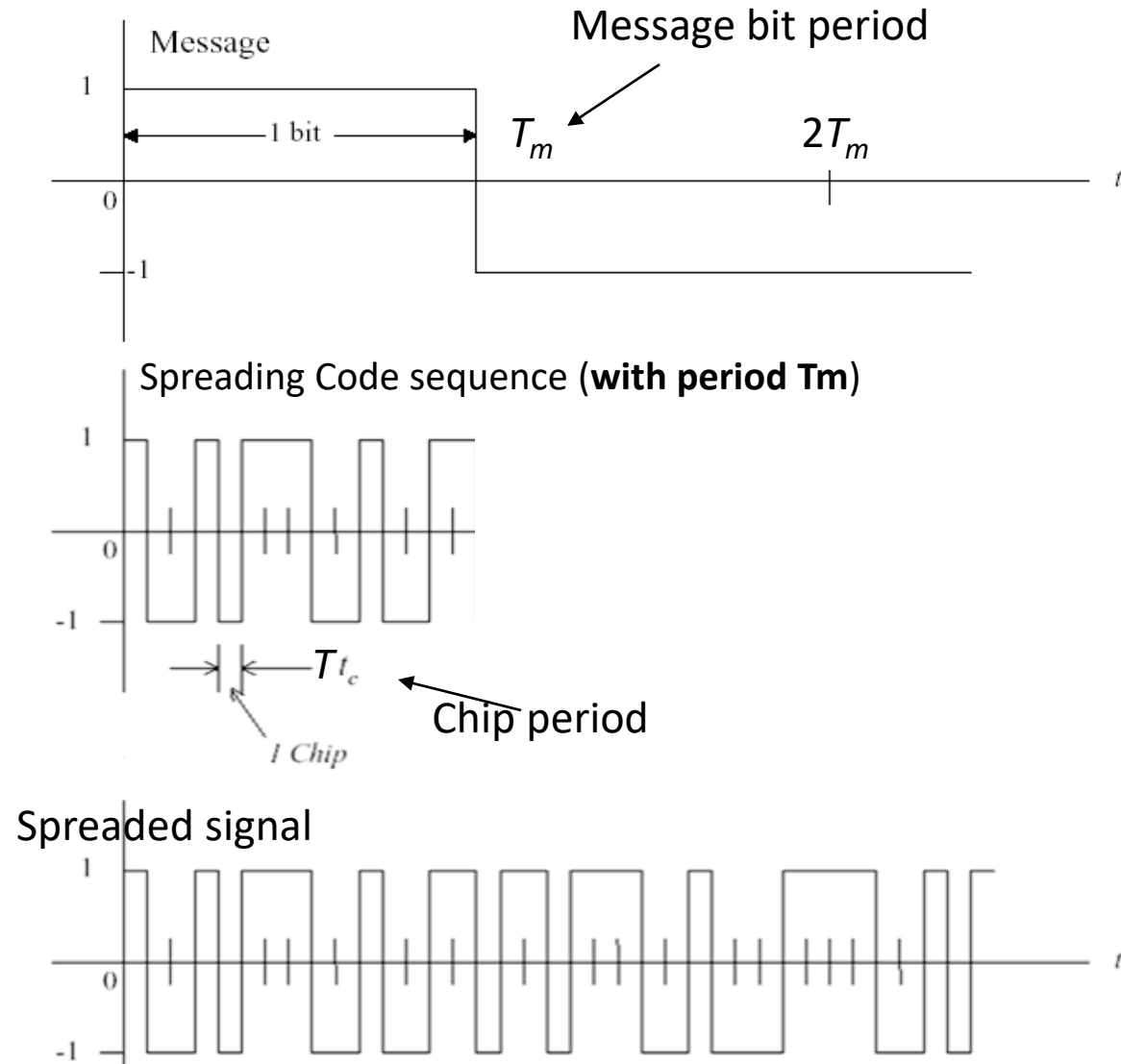
Direct Sequence (DS)-Spread Spectrum (SS): a modulation technique used in secure communication and 3G.



1.2 Signal Classification and Typical Signals



DS-SS

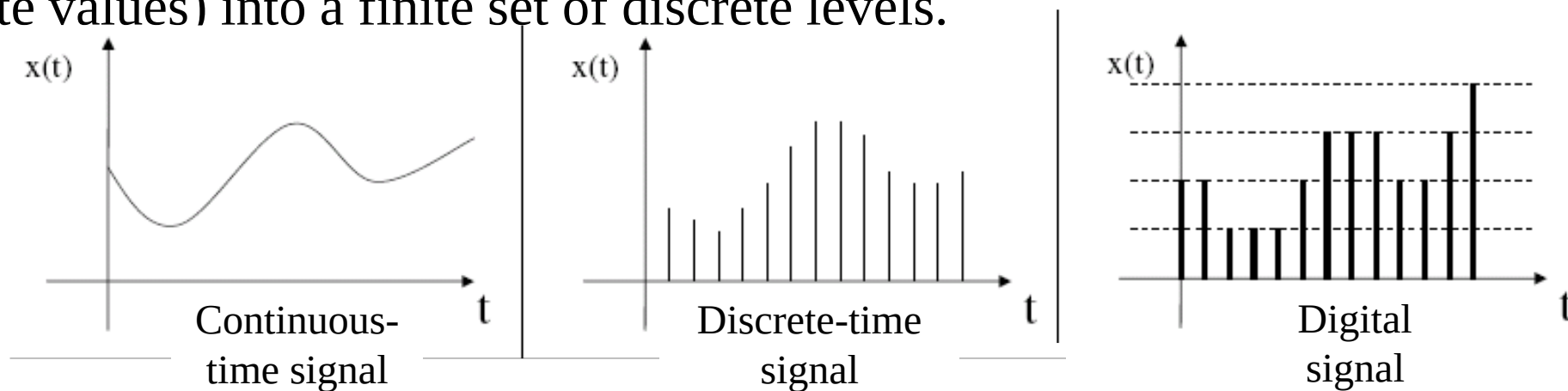


3. Continuous-Time Signals and Discrete-Time Signals

Continuous-time (analog) signal: Within the time interval under discussion, a definite function value can be given for any time value (except for several discontinuous points).

Discrete-time signal: The signal has values only at discrete time points and is undefined at other times. It is usually obtained via sampling of continuous-time signals.

Digital signal: The signal is discrete in both time and amplitude. **Quantization** is the process of converting a continuous range of analog signal amplitudes (or highly precise discrete values) into a finite set of discrete levels.



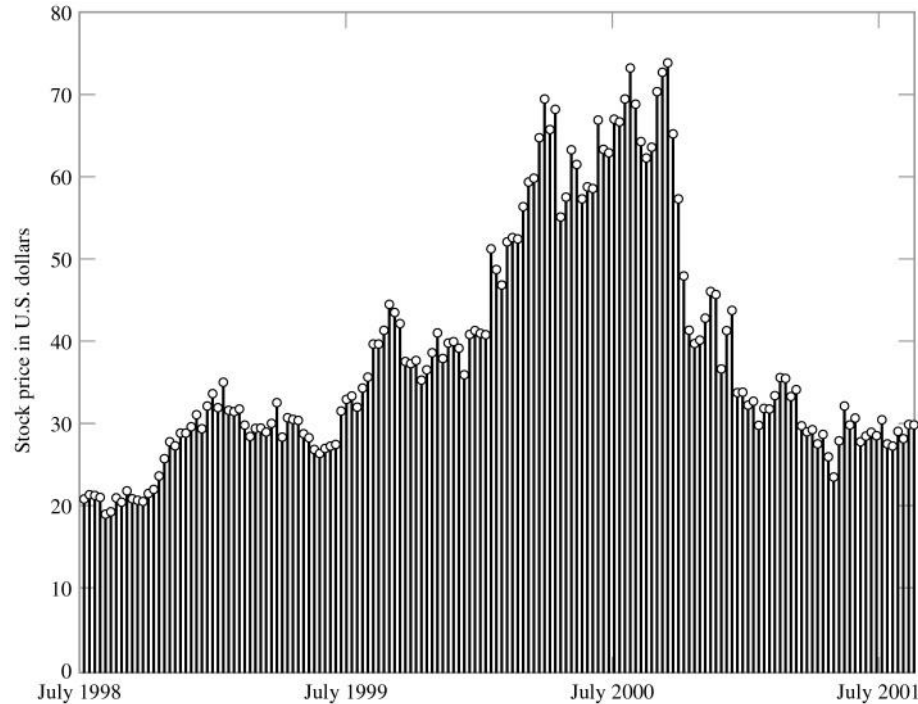
1.2 Signal Classification and Typical Signals



4. One-dimensional signals and multi-dimensional signals

A signal can be expressed as a function of one or more variables.

In this course, we mainly study **one-dimensional** signals where the variable is **time**.



Stock variation signal—one dimensional signal



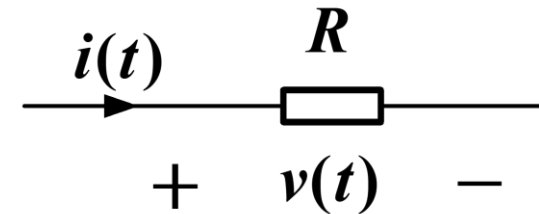
Picture—two-dimensional signal

5. Energy signals and power signals

Let $i(t)$ be the current through resistor R , and $v(t)$ be the voltage across resistor R .

The instantaneous power is

$$p(t) = i^2(t)R$$



The energy consumed by R over a time period is

$$E = \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} p(t) dt = R \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} i^2(t) dt \text{ 或 } E = \frac{1}{R} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} u^2(t) dt$$

The average power is

$$P = \frac{1}{T_0} R \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} i^2(t) dt \text{ 或 } P = \frac{1}{T_0} \frac{1}{R} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} u^2(t) dt$$

1.2 Signal Classification and Typical Signals



Definition: Generally, energy is always proportional to the square of a certain physical quantity.

Let $R = 1$, then within the entire time domain, for a real signal $f(t)$,

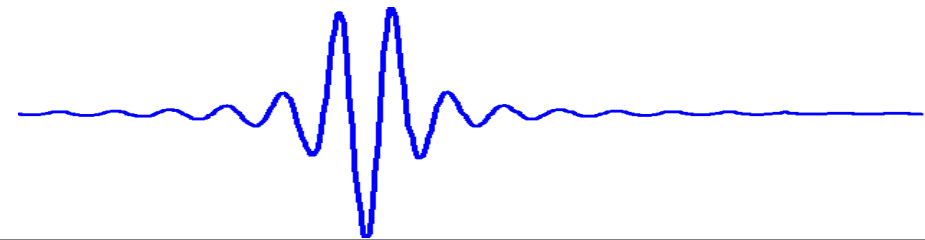
Energy:
$$E = \lim_{T_0 \rightarrow \infty} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f^2(t) dt$$

Average power:
$$P = \lim_{T_0 \rightarrow \infty} \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f^2(t) dt$$

There are two possibilities:

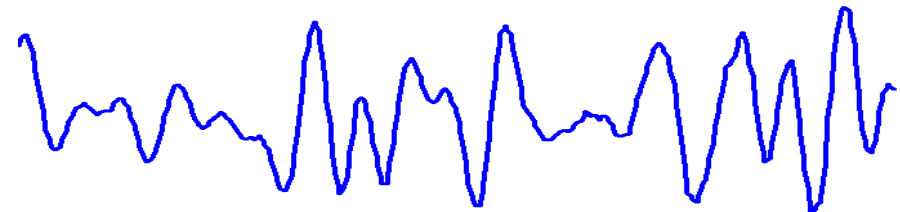
① $0 < E < \infty$ (with finite value),

$P = 0$ ← **Energy signal**



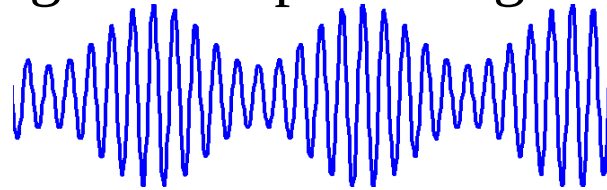
② $0 < P < \infty$ (with finite value),

$E = \infty$ ← **Power signal**

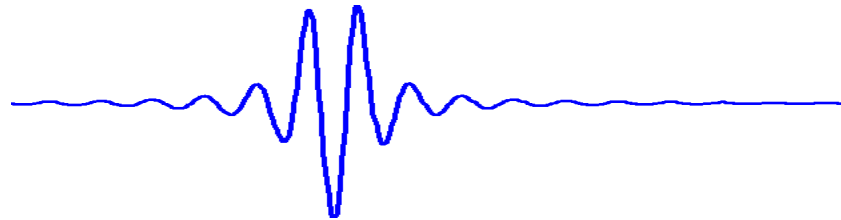


General Rules

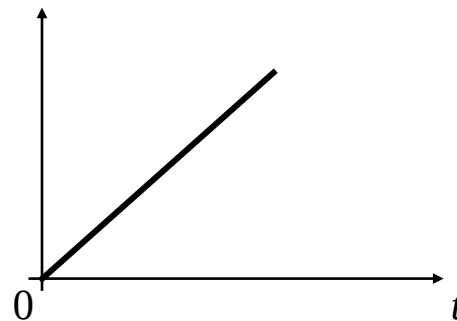
- ① Generally, periodic signals are power signals.



- ② Non-periodic signals with values within a limited interval are energy signals.



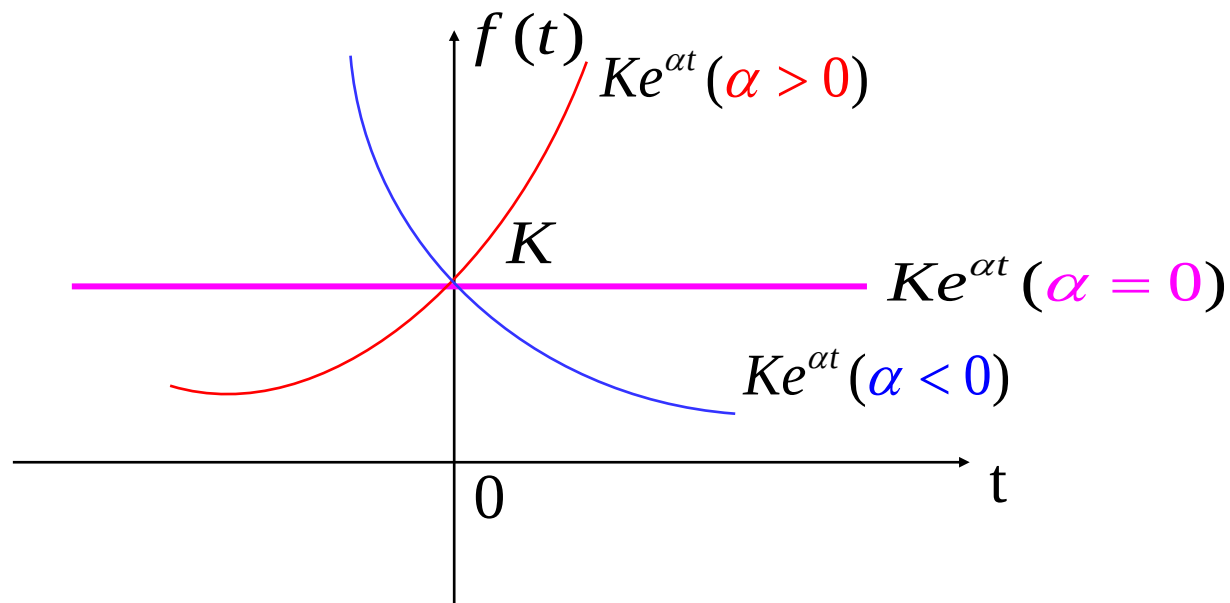
- ③ Some non-periodic signals are neither energy signals nor power signals.



1.2.2 Typical Signals

1. Exponential Signal

Expression: $f(t) = Ke^{\alpha t}$



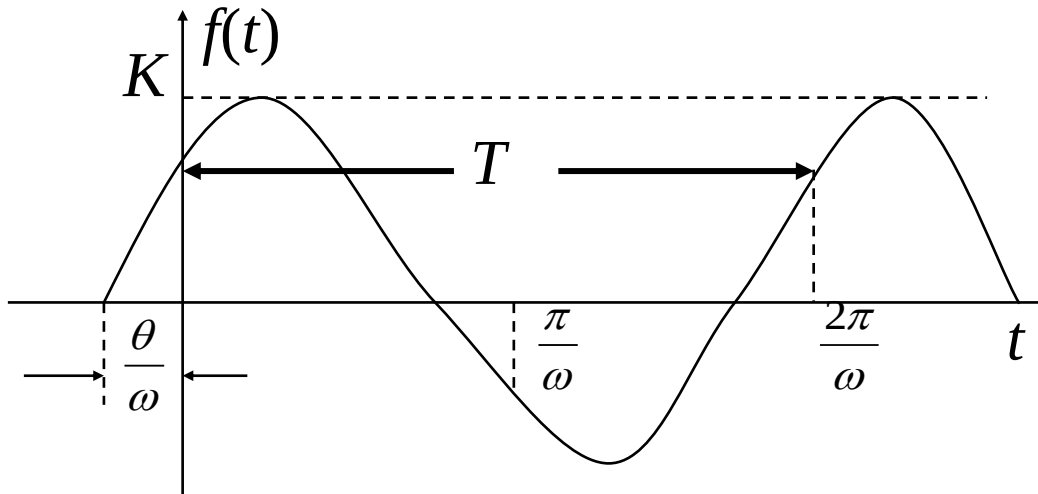
The result of differentiating or integrating an exponential function with respect to time is still in exponential form.

1.2 Signal Classification and Typical Signals



2. Sinusoidal Signal

$$f(t) = K \sin(\omega t + \theta)$$



$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$

$$e^{-j\omega t} = \cos \omega t - j \sin \omega t$$

$$\sin \omega t = \frac{1}{2j} (e^{j\omega t} - e^{-j\omega t})$$

$$\cos \omega t = \frac{1}{2} (e^{j\omega t} + e^{-j\omega t})$$

3. Complex Exponential Signal

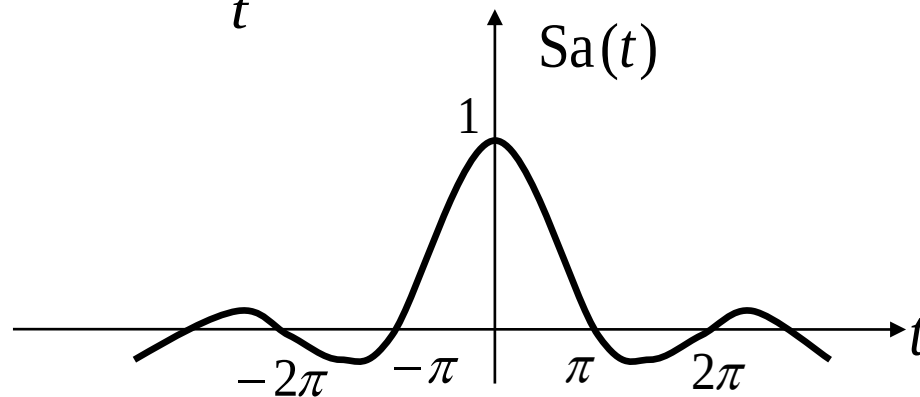
If the exponential factor of an exponential signal is a complex number, it is referred to as a complex exponential signal, expressed as:

$$f(t) = Ke^{st} = Ke^{(\sigma + j\omega)t} = Ke^{\sigma t} \cos \omega t + jKe^{\sigma t} \sin \omega t$$

4. Sampling function $\text{Sa}(t)$ and $\text{Sinc}(t)$

The sampling function is defined as:

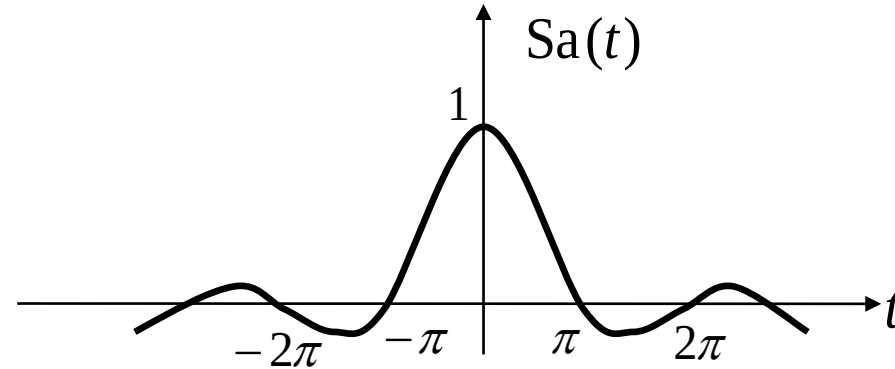
$$\text{Sa}(t) = \frac{\sin t}{t}$$



An alternative function is defined as:

$$\text{Sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

1.2 Signal Classification and Typical Signals



Properties of $Sa(t)$

(1) $Sa(t)$ is an even function, and its amplitude gradually decays in both the positive and negative directions of t .

$$(2) \quad \int_0^{\infty} Sa(t)dt = \frac{\pi}{2} \qquad \int_{-\infty}^{\infty} Sa(t)dt = \pi$$

1.3.1 Addition of Signals

The sum (or difference) of two signals is still a signal, and its value at any time is equal to the sum (or difference) of the values of the two signals at that time, i.e.,

$$f(t) = f_1(t) + f_2(t) \quad \text{or} \quad f(t) = f_1(t) - f_2(t)$$

1.3.2 Multiplication and Scaling of Signals

The product of two signals is still a signal, and its value at any time is equal to the product of the values of the two signals at that time, i.e.,

$$f(t) = f_1(t) f_2(t)$$

The scaling operation of a signal refers to multiplying a signal by a real constant K , which means multiplying the value of the original signal at each time by K , i.e.,

$$f(t) = Kf(t)$$

1.3 Signal Operations

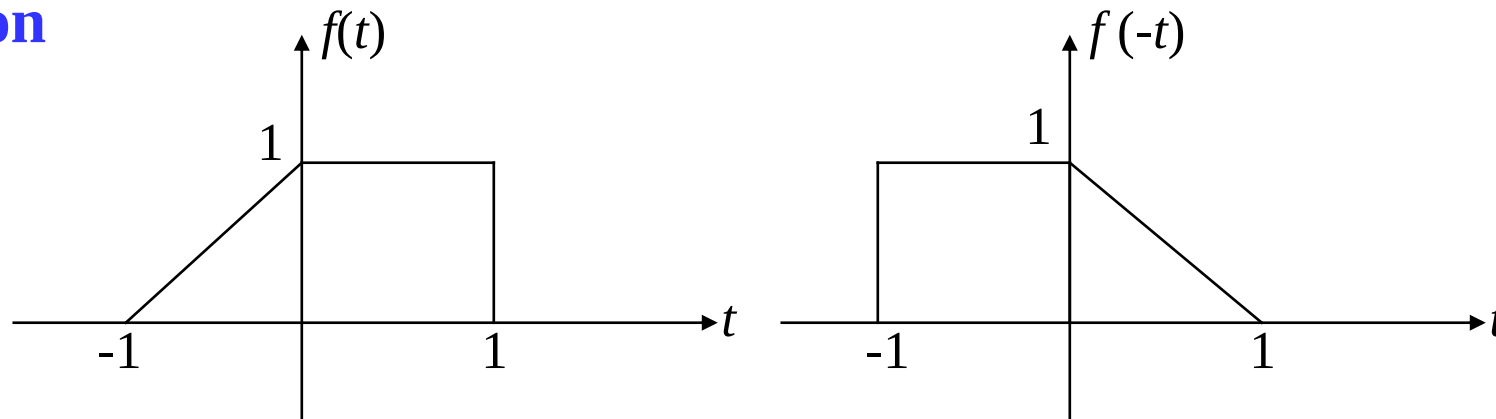


1.3.3 Time Operations of Signals-Reverse, Time Shifting and Scaling

1. Reverse Operation

$$f(t) \rightarrow f(-t)$$

Reverse against the
 $t = 0$ axis.

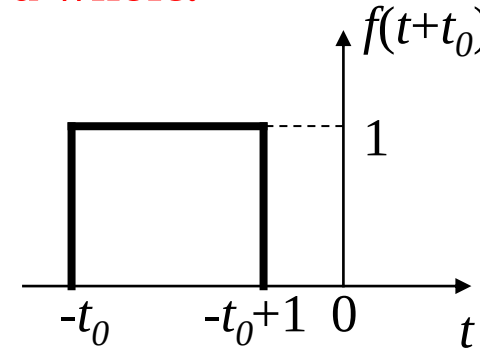
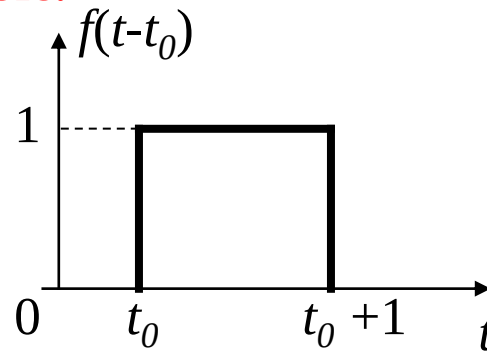
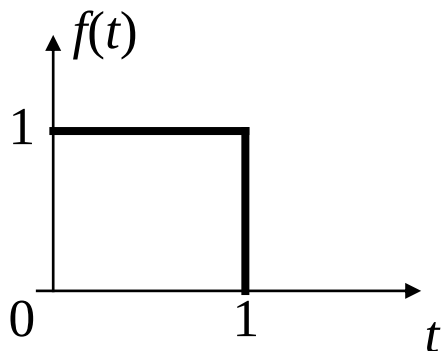


2. Time Shifting Operation

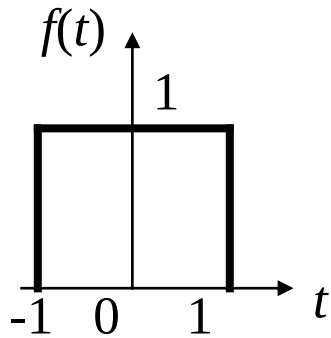
$$f(t) \rightarrow f(t - t_0)$$

$t_0 > 0$ 时, $f(t)$ shifts to the right by
 t_0 as a whole.

$t_0 < 0$ 时, $f(t)$ shifts to the left by
 t_0 as a whole.

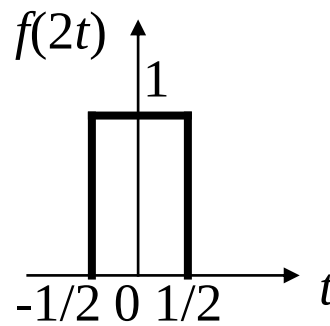


3. Scaling Operation



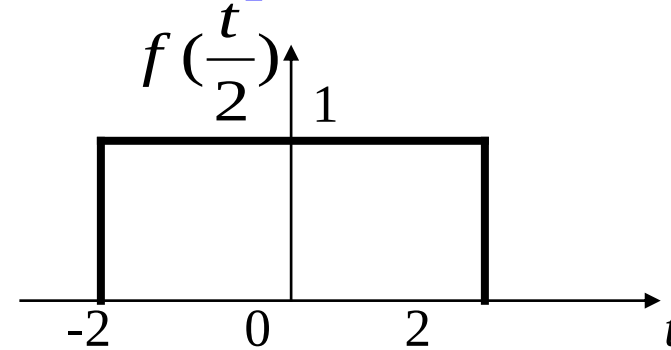
$$f(t) \rightarrow f(2t)$$

Compression



$$f(t) \rightarrow f\left(\frac{t}{2}\right)$$

Expansion



$$f(t) \rightarrow f(at)$$

Compression: If $|a| > 1$, the waveform is compressed along the time axis.

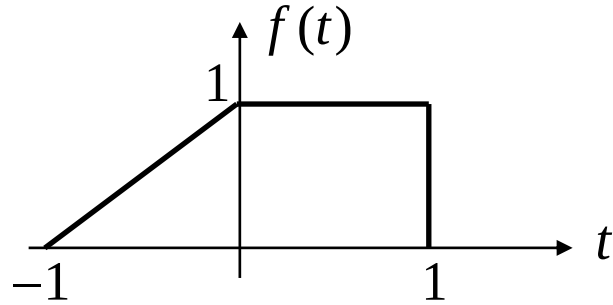
Expansion: If $0 < |a| < 1$, the waveform is expanded along the time axis.

Please give one real-life example each for signal reverse, time shifting and compression.

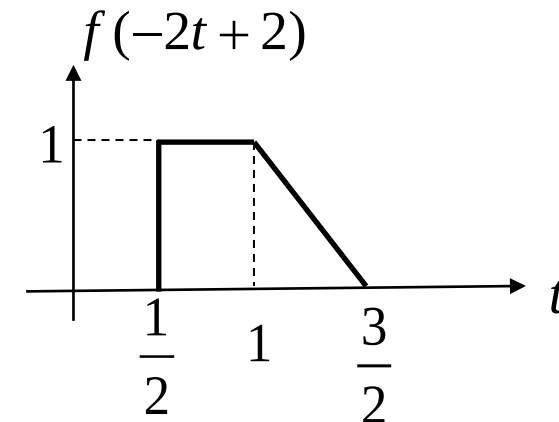
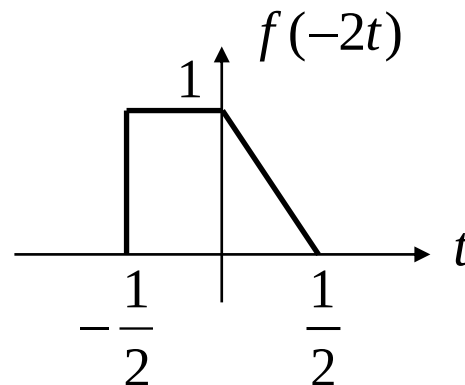
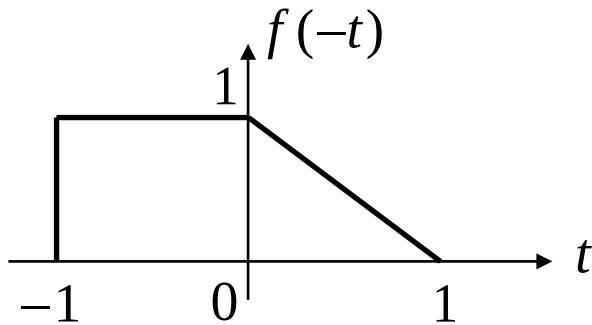
1.3 Signal Operations



Example 1-1: The signal is shown in the figure below, find $f(-2t+2)$, and sketch the waveform.



$$f(t) \xrightarrow{\text{reverse}} f(-t) \xrightarrow{\text{scaling}} f(-2t) \xrightarrow{\text{shift}} f(-2t+2) = f[-2(t-1)]$$

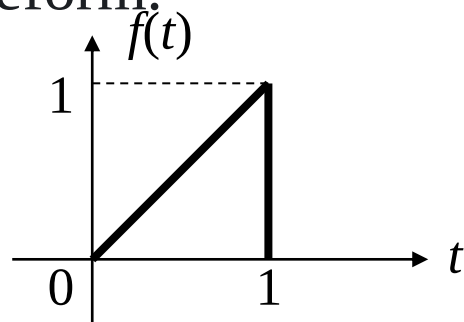


1.3.4 Differentiation and Integration Operations of Signals

1. Differentiation Operation

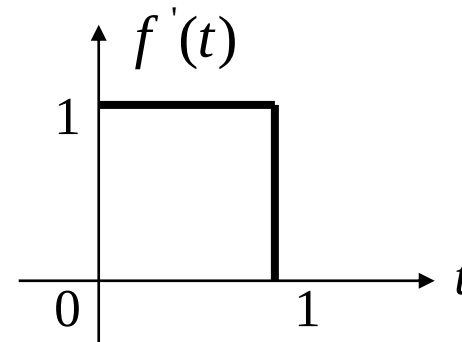
The derivative of signal $f(t)$ is still a signal, represented by $f'(t)$, which represents the rate of change of the signal with time. The differentiation operation highlights the changing part of the signal.

Example 1-2: Find the derivative $f'(t)$ of signal $f(t)$ shown in the figure below and sketch its waveform.



Solution: $f(t) = t \quad (0 < t \leq 1)$

$$f'(t) = 1 \quad (0 < t \leq 1)$$



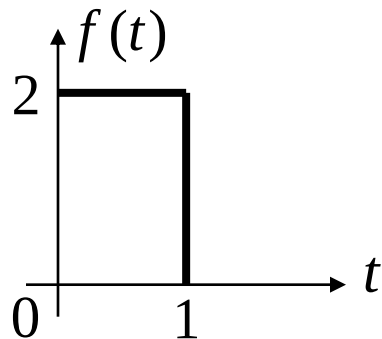
2. Integration Operation

- The integral of signal $f(t)$, denoted by $\int_{-\infty}^t f(\tau) d\tau$ or $f^{(-1)}(t)$, is still a signal. Its value at any time is equal to the area enclosed by $f(t)$ and the time axis within the interval from $-\infty$ to t .
- **The integration operation smooths the abrupt parts of the signal and can reduce the impact of glitches (noise).**

1.3 Signal Operations



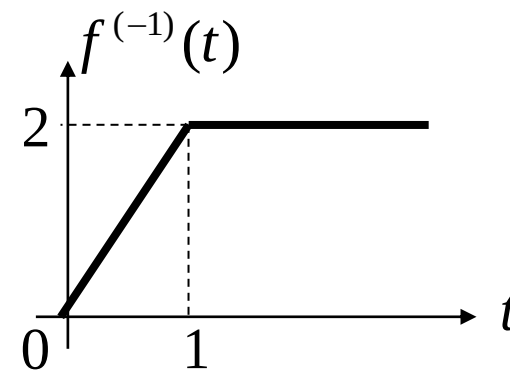
Example 1-3: Select the waveform of the integral $f^{(-1)}(t) = \int_{-\infty}^t f(\tau) d\tau$ of signal $f(t)$ shown in the figure.



Solution : 1) If $t < 0$ 时, $f^{(-1)}(t) = 0$

2) If $0 \leq t \leq 1$ 时, $f^{(-1)}(t) = \int_0^t 2 d\tau = 2t$

3) If $t > 1$ 时, $f^{(-1)}(t) = \int_0^1 2 d\tau = 2$

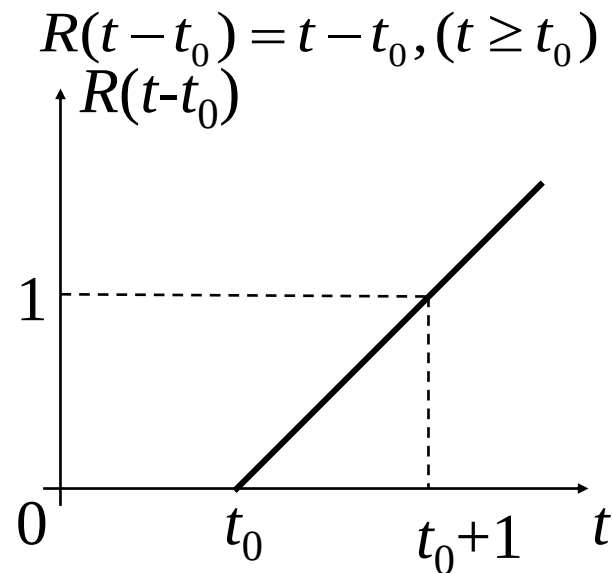
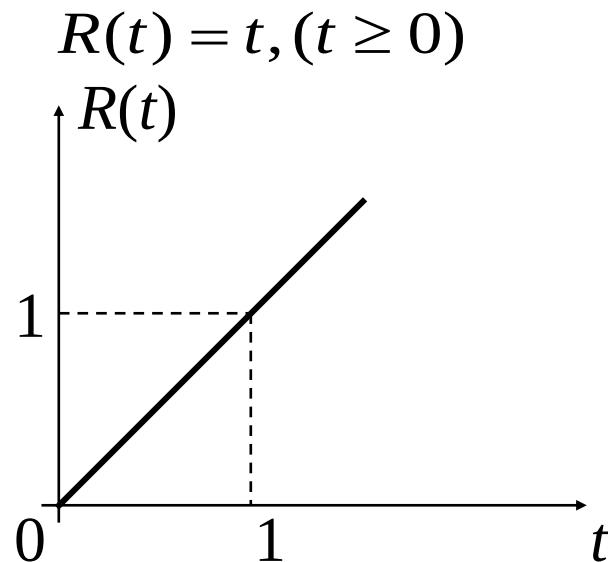


1.4 Singular Signals

In the analysis of signals and systems, we often encounter functions with discontinuities in themselves or their derivatives/integrals. Such functions are collectively called singular functions or singular signals.

1.4.1 Unit Ramp Signal

A unit ramp signal refers to a signal that increases proportionally with a unit slope, starting from time 0.

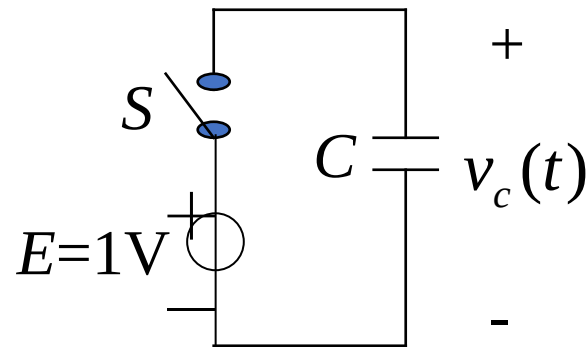


1.4.2 Unit Step Signal

$$u(t) = \begin{cases} 0, & (t < 0) \\ 1, & (t > 0) \end{cases}$$



What is the physical meaning of the unit step signal in circuits?

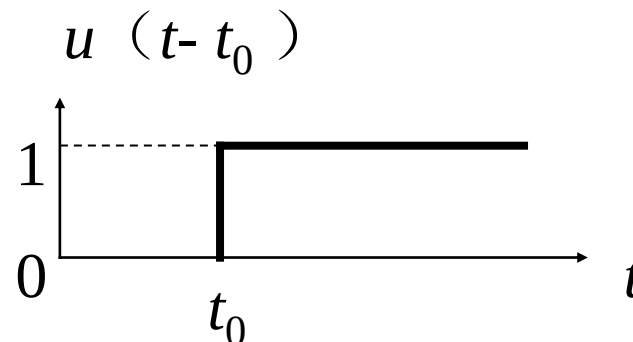


Example 1-4: Assume that S , E , and C in the figure are all ideal components (internal resistance = 0). Find the voltage across capacitor C when S is closed at $t = 0$.

Solution: Since S , E , and C are all ideal components, there is no internal resistance in the circuit. After S is closed, the voltage across C will jump, forming a step voltage. That is:

$$v_c(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases} = u(t)$$

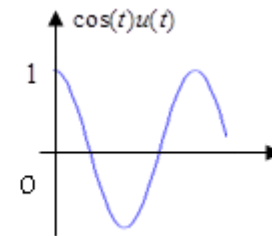
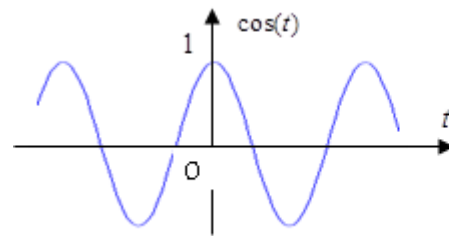
If switch S is closed at $t = t_0$, the voltage across the capacitor is $u(t - t_0)$. The waveform is shown below.



Properties of $u(t)$: Unilateral characteristic, i.e.,

$$f(t)u(t) = \begin{cases} 0 & t < 0 \\ f(t) & t > 0 \end{cases}$$

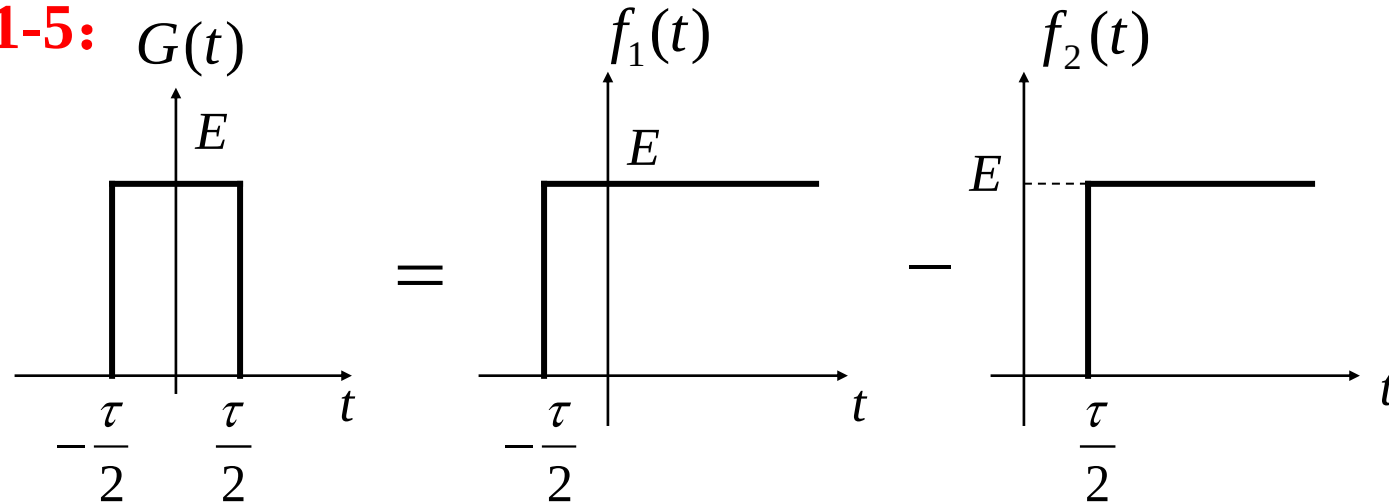
Some pulse signals can be expressed using step signals.



1.4 Singular Signals



Example 1-5:



$$\text{Since } f_1(t) = Eu(t + \frac{\tau}{2}), \quad f_2(t) = Eu(t - \frac{\tau}{2}),$$

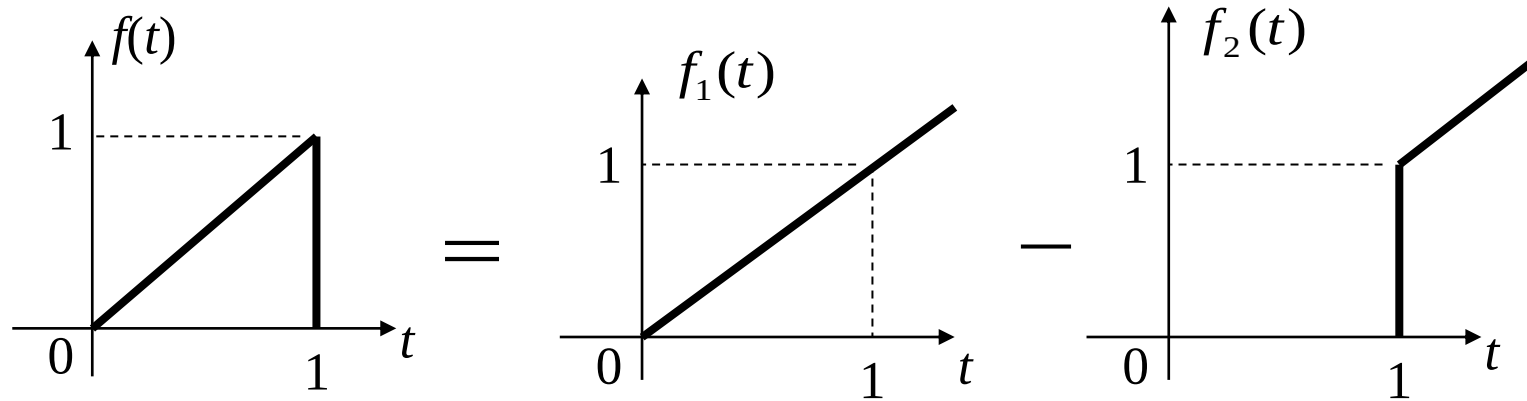
the rectangular pulse $G(t)$ can be expressed as:

$$G(t) = f_1(t) - f_2(t) = E[u(t + \frac{\tau}{2}) - u(t - \frac{\tau}{2})]$$

1.4 Singular Signals



Example 1-6:



$$f(t) = t[u(t) - u(t-1)]$$

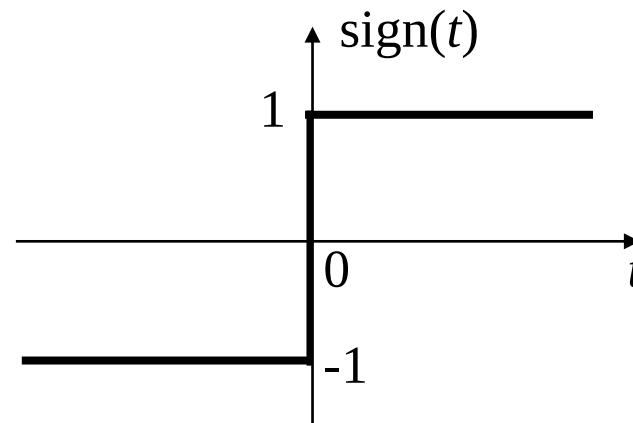
Example 1-7: Which of the following is the correct expression for the "signum function" ($\text{sign}(t)$) using step signals?

☐ A $1 - 2u(t)$

☒ B $2u(t) - 1$

☐ C $2u(t) - 2$

☐ D $2 - 2u(t)$

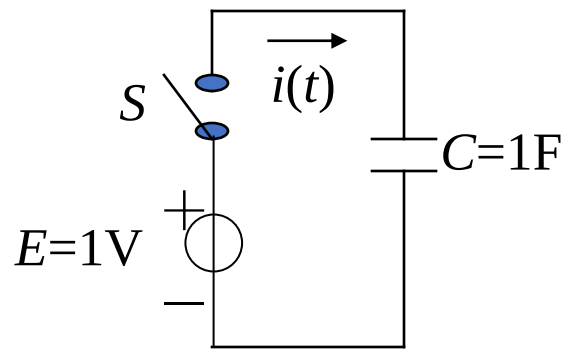


1.4 Singular Signals



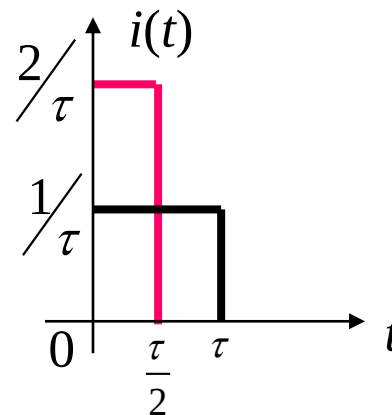
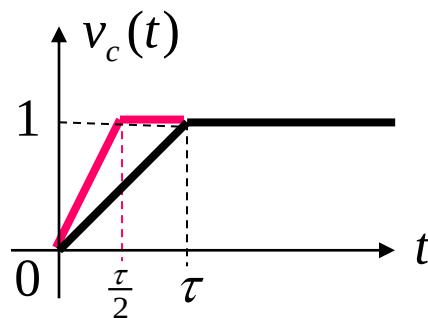
1.4.3 Unit Impulse Signal (Dirac Delta Function)

Physical meaning of the unit impulse signal (Dirac delta function, $\delta(t)$) with the following example:

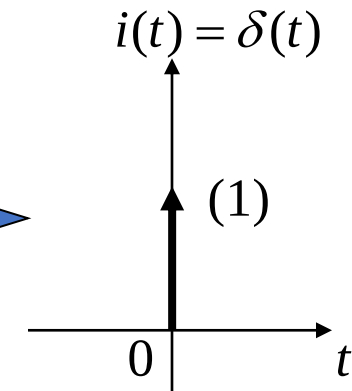


Assume that S, E, and C in the figure are all ideal components (internal resistance = 0). Find current $i(t)$ when S is closed at $t = 0$.

$$i(t) = C \frac{dv_c(t)}{dt}$$



$\tau \rightarrow 0$



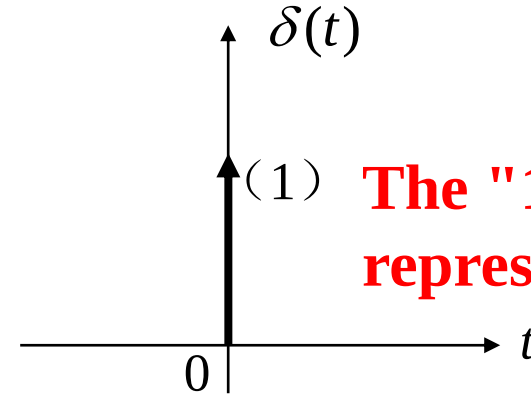
1.4 Singular Signals



1. Definition of $\delta(t)$

1) By Expression

$$\begin{cases} \delta(t) = 0 & (t \neq 0) \\ \int_{-\infty}^{\infty} \delta(t) dt = 1 \end{cases}$$

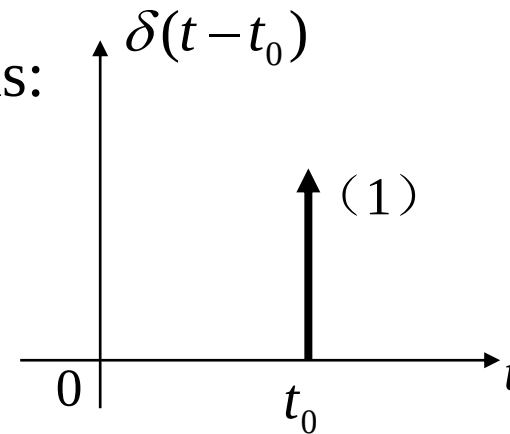


The "1" in the bracket represents the strength.

This definition method was proposed by Dirac.

Similarly, $\delta(t - t_0)$ can be defined as:

$$\begin{cases} \delta(t - t_0) = 0 & (t \neq t_0) \\ \int_{-\infty}^{\infty} \delta(t - t_0) dt = 1 \end{cases}$$



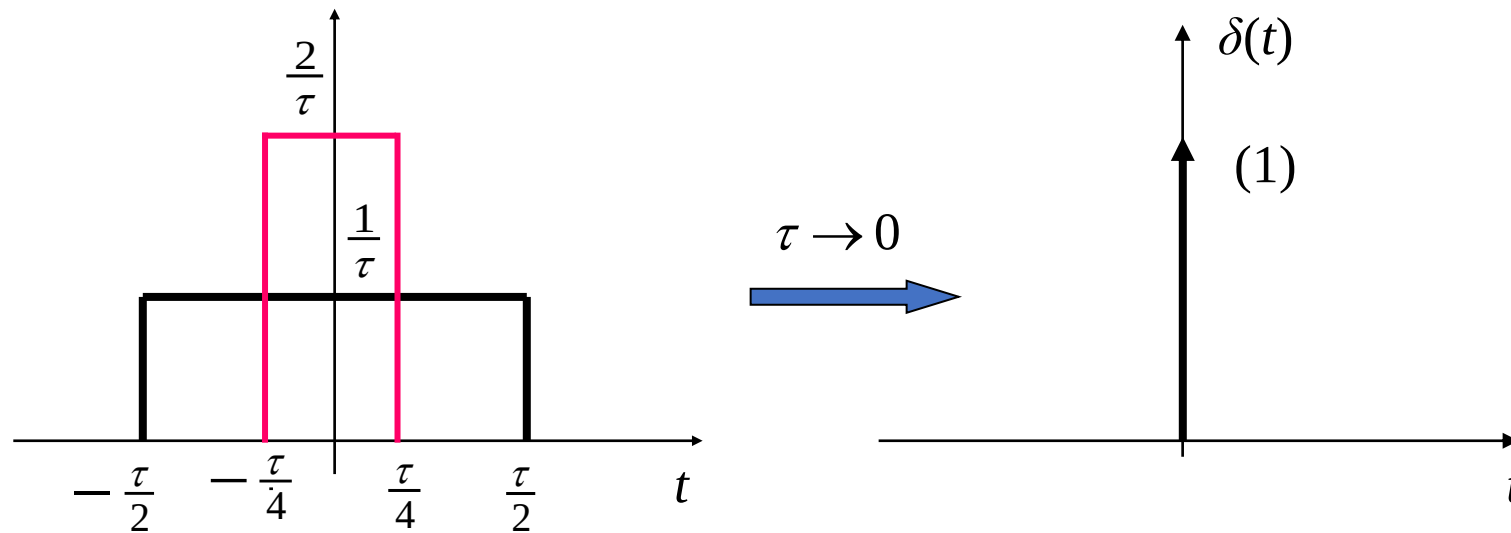
1.4 Singular Signals



2) By Limit

We can define $\delta(t)$ by taking the limit of various regular function sequences.

For example, (a) definition by the limit of rectangular pulses

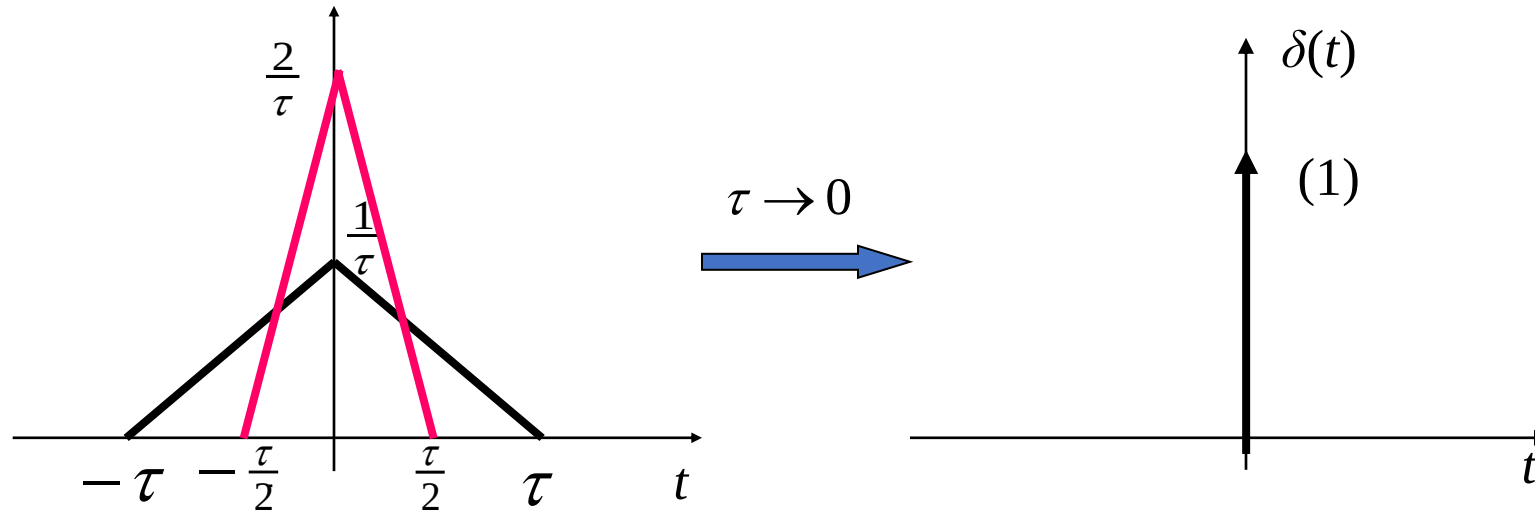


$$\delta(t) = \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$$

1.4 Singular Signals



(b) Definition by the limit of triangular pulses



$$\delta(t) = \lim_{\tau \rightarrow 0} \left\{ \frac{1}{\tau} \left(1 - \frac{|t|}{\tau} \right) [u(t + \tau) - u(t - \tau)] \right\}$$

2. Properties of the Unit Impulse Signal

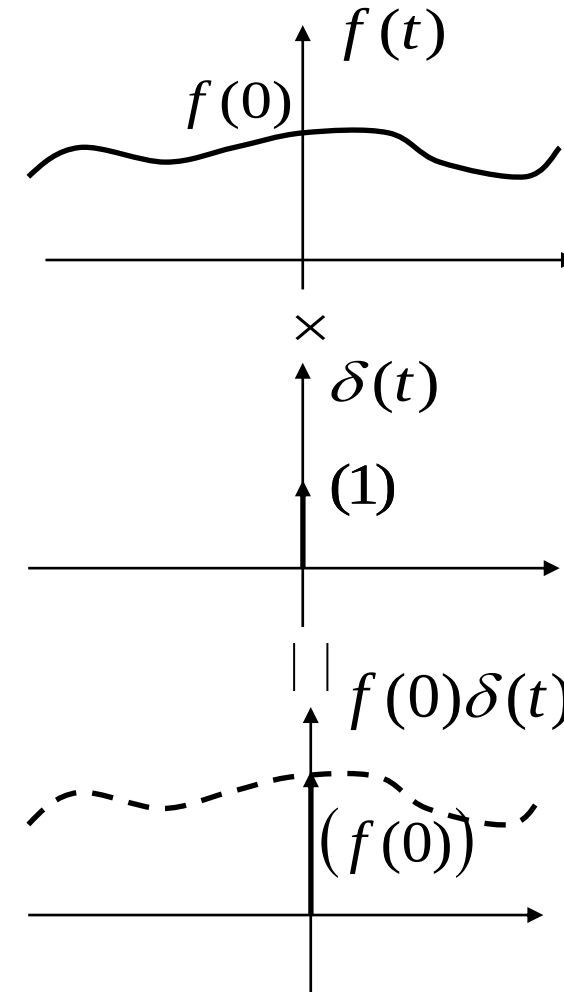
1) Sampling

$$f(t)\delta(t) = f(0)\delta(t) \quad (1)$$

$$\int_{-\infty}^{\infty} \delta(t)f(t)dt = f(0)\int_{-\infty}^{\infty} \delta(t)dt = f(0) \quad (2)$$

$$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0) \quad (3)$$

$$\int_{-\infty}^{\infty} \delta(t-t_0)f(t)dt = f(t_0) \quad (4)$$



Combining formulas (2) and (4), we can draw the following conclusion:

The impulse function can extract (sample) the function value at the position where the impulse is located.

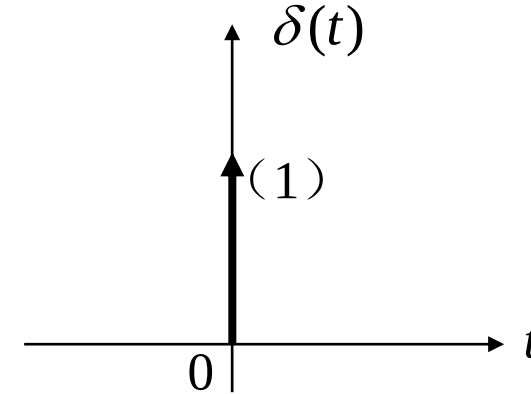
1.4 Singular Signals



2) $\delta(t)$ is an even function, i.e., $\delta(t) = \delta(-t)$

$$3) \int_{-\infty}^t \delta(\tau) d\tau = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases} = u(t)$$

$$\int_{-\infty}^t \delta(\tau - t_0) d\tau = u(t - t_0)$$



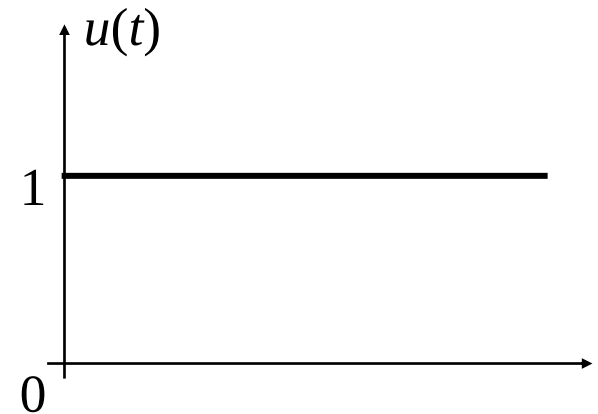
Relational ship between $u(t)$ and $\delta(t)$

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t)$$

$$\frac{d}{dt} u(t) = \delta(t)$$

$$\int_{-\infty}^t \delta(\tau - t_0) d\tau = u(t - t_0)$$

$$\frac{d}{dt} u(t - t_0) = \delta(t - t_0)$$

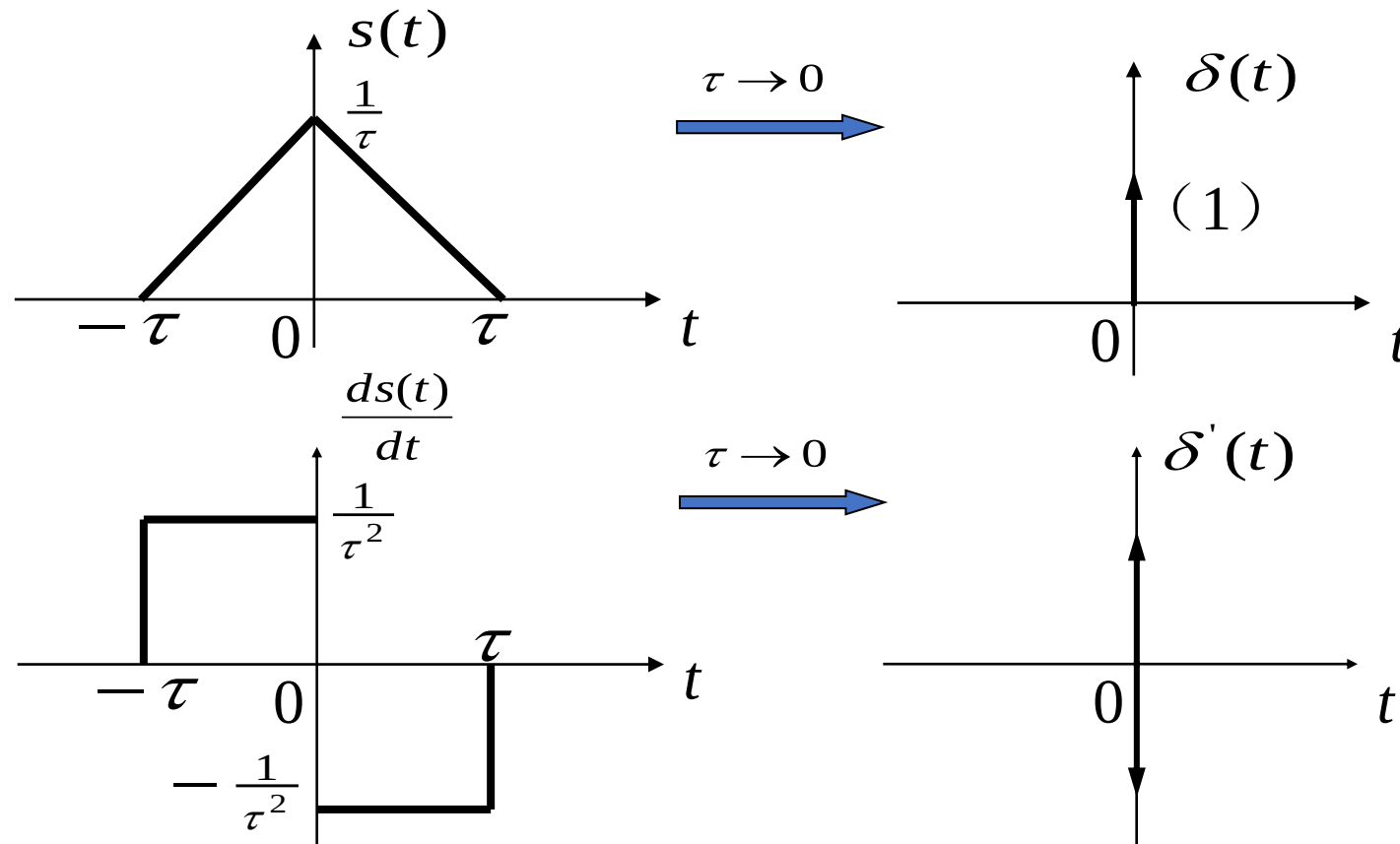


1.4 Singular Signals



1.4.4 Impulse Derivative (Doublet)

The derivative of the impulse function (second derivative of the step function) presents a pair of impulses with positive and negative polarities, which is called the impulse derivative (doublet), denoted by $\delta'(t)$.



Properties of the Impulse Derivative

1) The impulse derivative is an odd function, i.e., $\delta'(-t) = -\delta'(t)$

$$2) \quad \int_{-\infty}^{\infty} \delta'(t) f(t) dt = -f'(0)$$

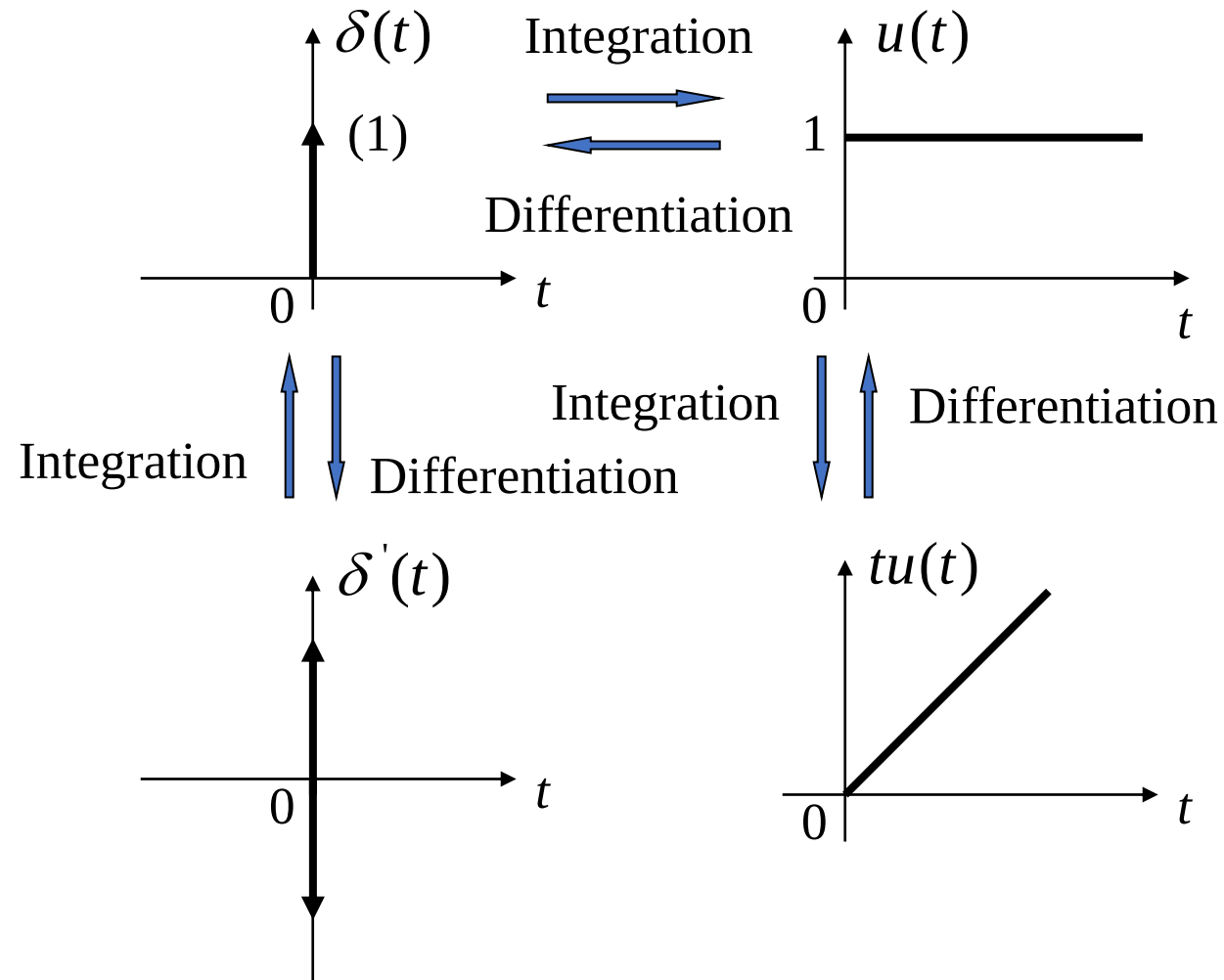
$$\int_{-\infty}^{\infty} \delta'(t-t_0) f(t) dt = -f'(t_0)$$

$$3) \quad \int_{-\infty}^{\infty} \delta'(t) dt = 0$$

1.4 Singular Signals



Relationship between $tu(t)$, $u(t)$, $\delta(t)$, and $\delta'(t)$:



- Homework: 1.50, 1.51, and 1.57(c)(f)(g) in the textbook.
- Simon Haykin and Barry Van Veen, Signals and Systems (Second Edition), John Wiley & Sons, 2002, 9780471164746.
- You do not have to submit the homework, but you are highly recommended to do the homework. Refer to the solutions after you finish the exercises.